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Prediction of exposure to pollutants and hormones on the risk of polycystic ovarian syndrome

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Abstract

Background There has been little research on the association of exposure to environmental factors on polycystic ovary syndrome (PCOS), nor on the interaction between environmental factors and liver and kidney function. Anti-mullerian hormone (AMH) has been proposed to add significance to diagnosis of PCOS in case of ambiguity. We hypothesize that long-term inhalation exposure to environmentally relevant levels of these factors may induce changes in hepatic and renal function, thereby exacerbating the risk of developing PCOS.

Methods The study used a cross-sectional study. Cases were newly diagnosed PCOS patients from a tertiary hospital. Controls were age - and BMI - matched healthy women recruited from the same communities. Data on age and various blood test results were collected from medical records. Meteorological factors and air pollutants were obtained from the National Oceanic and Atmospheric Administration (NOAA). After feature selection, we employed logistic regression, weighted quantile sum (WQS) regression, and neural network models to analyze the associations between relevant variables and the risk characteristics and prediction of PCOS including different aged groups.

Results There were 384 subjects in this retrospective study, randomly including 178 PCOS patients and 206 controls. The levels of most sexual function (FSH, LH, PRL, T, AMH) and liver function indicators (TP, Alb, A/G, ALP, PA, TBA) in PCOS patients were significantly higher than those in the control group. Overall, the AMH level in the PCOS population was 1.133 times that of the non-affected population (95% confidence interval [CI]: 1.077, 1.192). Within the 21–35 years age group, the levels of air pressure and albumin in PCOS patients were 1.060 (95% CI: 1.028, 1.093) and 1.098 (95% CI: 1.002, 1.204) times higher, respectively, than in the non-affected population. Based on the results obtained from the stratified analysis, we incorporated several variables into the prediction model, namely PM_{2.5}, air pressure, FSH, PRL, T, AMH, Alb and PA. The overall population demonstrated good PCOS predictive performance in internal validation using the neural network model (test AUC = 0.864, train AUC = 0.992; test R² = 0.342, train R² = 0.910).

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Conclusions Significant elevations in levels of AMH and Alb were detected in women with PCOS. The back-propagation (BP) neural network demonstrated good PCOS predictive performance for the models mediated by environmental factors (PM_{2.5}, air pressure). This suggests that these factors may probably exacerbate the effects of sexual function (FSH, PRL, T, AMH) and liver function indicators (Alb, PA) on the risk of developing PCOS. Our results support a potential association between environmental factor exposure and the consequences of PCOS in women.

Keywords Polycystic ovary syndrome, Environmental exposure, BP neural network, AMH, Mediating effect

Introduction

Polycystic Ovary Syndrome (PCOS) was initially described as a reproductive disorder characterized by enlarged, smooth polycystic ovaries, irregular menstrual cycles, infertility, and hirsutism [1]. Although studies from Greece, Spain, and the United States have traditionally estimated the prevalence of PCOS has traditionally been estimated at 4%-8% [2], recent research found that 70% of women remain undiagnosed [3]. In 2006, based on U.S. data and traditionally lower prevalence estimates, the expected economic burden of PCOS in Australia was AUD 400 million (with PCOS-related diabetes accounting for 40% of the total costs), highlighting its significant health and economic impact [4]. A recent study indicated that 24 million reproductive-aged women in China are affected by PCOS, with the estimated prevalence in 2020 being higher than a decade ago [5]. These findings underscore PCOS as a major public health concern.

PCOS is characterized by a series of interrelated changes in reproductive hormones, such as the release of follicle-stimulating hormone (FSH), enhanced luteinizing hormone (LH), and elevated levels of anti-Müllerian hormone (AMH). AMH is secreted by the granulosa cells of preantral and small antral follicles and serves as a regulatory molecule in ovarian folliculogenesis. Circulating AMH levels are proportional to the number of growing ovarian follicles. Consequently, AMH levels are increased in PCOS and are considered a marker of the unique follicular dysfunction associated with the syndrome [6, 7]. Genome-wide association studies (GWAS) have identified several specific genetic susceptibility loci in women with PCOS [8], with 11 loci found in women of both Chinese and European ancestries. This suggests a common genetic basis for the disease across different ethnic groups. However, the loci identified from GWAS account for less than 10% of the heritability of PCOS, indicating that socioeconomic, cultural, or environmental factors may also contribute to its development. A study in Shanghai, China, showed that increased levels of PM₁₀ were associated with a decreased clinical pregnancy rate in women without PCOS, but no significant correlation was found between air pollutants and patients with PCOS [9]. A recent review study from Harvard University suggests that environmental exposure is an important emerging area in menstrual research. The studies on environmental exposure and PCOS are limited, and the

results across different studies show heterogeneity [10]. For example, a review study from Australia concluded that fine particulate air pollution may increase the risk of PCOS, but the existing evidence remains inconsistent [11]. Meanwhile, the latest research in our country has found that PMs exposure is associated with a decrease in cysteine levels in follicular fluid, but further associations with the disease are still lacking [12]. This highlights the hope that future research will help fill the gap in understanding how environmental exposure affects PCOS, and will be of significant importance in confirming the relationship between environmental exposure and PCOS in this study.

This study aims to compare the differences in sex hormones and liver and kidney function between patients with PCOS and non-PCOS controls. Additionally, we seek to evaluate the correlation between PCOS and environmental factors, and to establish a predictive model for PCOS. We will also assess whether there is a mediating effect of environmental factors on the association between blood markers and PCOS.

Patients and methods

Study population and validation process

This study was approved by the Ethics Committee of Shengjing Hospital affiliated with China Medical University (ID: 2024PS123K). The studies were conducted in accordance with the Declaration of Helsinki, local legislation and institutional requirements. Patients were recruited from the obstetrics and gynecology outpatient department of Shengjing Hospital from April 2020 to May 2023. Cases were newly diagnosed PCOS patients from a tertiary hospital. Controls were age- and BMI-matched healthy women recruited from the same communities. Data on cases reports, age and various blood test results were collected from medical records. Comprehensive demographic and health data were collected from 178 patients with PCOS and 206 control subjects at the medical center of Shengjing Hospital affiliated with China Medical University. The diagnosis of PCOS in women in this study was based on the 2003 Rotterdam criteria, which require the presence of at least two of the following three features: (1) oligo-ovulation and/or anovulation; (2) clinical and/or biochemical signs of hyperandrogenism; and (3) polycystic ovaries. Exclusion criteria included exclusion of other causes of

hyperandrogenism or ovulatory dysfunction, such as Cushing's syndrome, 21-hydroxylase deficiency, congenital adrenal hyperplasia, androgen-secreting tumors, and hyperprolactinemia [13]. The control participants consisted of healthy women without endocrine disorders. To ensure the accuracy of the control group, potential PCOS patients were excluded through ultrasound examination and serum hormone level assessment. None of the control group participants exhibited characteristics of PCOS according to the 2003 Rotterdam criteria (oligo/anovulation, clinical or biochemical signs of hyperandrogenism, and polycystic ovaries) [14].

Clinical and environmental characteristics

General information and laboratory tests were collected through case retrieval. General information included age, while laboratory monitoring parameters encompassed sex hormone levels (Follicle-Stimulating Hormone (FSH), Luteinizing Hormone (LH), Prolactin (PRL), Progesterone (P), Estradiol (E2), Testosterone (T), Sex Hormone-Binding Globulin (SHBG), Anti-Müllerian Hormone (AMH)), thyroid function indicators (Free Triiodothyronine (FT3), Free Tetraiodothyronine (FT4), Thyroid-Stimulating Hormone (TSH), Thyroid Peroxidase Antibody (TPOAb), Thyroglobulin Antibody (TGAb)), liver function tests (Total Protein (TP), Albumin (Alb), Albumin/Globulin Ratio (A/G), Aspartate Aminotransferase (AST), Alkaline Phosphatase (ALP), γ -Glutamyl Transferase (GGT), Prealbumin (PA), Total Bilirubin (TB), Conjugated Bilirubin (CB), Unconjugated Bilirubin (UCB), Total Bile Acid (TBA), Monoamine Oxidase (MAO)) and renal function indicators (urea, Creatinine (Cr), Uric Acid (UA)). Thyroid-related function indicators are mainly detected by chemiluminescent immunoassay, and the reference ranges for the general population are as follows: $FT3 \in [2.43, 6.01]$, $FT4 \in [9.01, 19.05]$, $TSH \in [0.30, 4.80]$, $TPOAb \in [0, 5.61]$, $TGAb \in [0, 4.11]$. Each participant was from the same geographic area (thus having the same exposure values). This study explored the association between environmental exposure and population health outcomes by analyzing the correlation between urban-scale environmental pollutant monitoring data and the incidence of specific diseases in regional hospitals. The unique advantages of the urban-scale study can precisely compensate for the limitations of individual-level research from a macroscopic perspective, which are specifically manifested as follows: (a) Advantage in preliminary verification of macroscopic association rules: Urban-scale studies can integrate spatiotemporal distribution data of pollutants across the entire region and big data on disease surveillance from hospitals, thereby quickly identifying the core association direction between environmental exposure and disease incidence. (b) Advantage in direct support for public

health decision-making: Compared with individual or community-scale studies, the results of correlation analysis at the urban scale are more in line with the needs of formulating urban-level public health policies. Meteorological factors and air pollutants, such as air pressure, CO, NO₂, O₃, etc., at the monitoring station nearest to the patient population, were obtained from the National Oceanic and Atmospheric Administration (NOAA) (<https://www.noaa.gov/>).

Weighted Quantile Sum (WQS) regression

We employed WQS regression model to investigate the importance of clinical and environmental factors in relation to the characteristics of PCOS patients. The WQS regression model is utilized to assess the confounding effects of multiple exposure variables on a given outcome variable, with each exposure variable assigned a weight in the model to indicate its degree of influence on the outcome variable [15].

The prediction quality of BP (back propagation) neural network

R software (version 4.1.3) was used in this study. To ensure that each model was consistently trained, each model was trained on 75% of the training sample. A common 25% hold set was retained for all models and used to produce statistics for comparing results between them. The predictive factors primarily included those factors that showed statistical significance in the previous multivariate regression analysis, which were incorporated into this model. BP neural network algorithms with different functions were applied to predict PCOS diseases. The BP neural network is a multilayer feedforward neural network trained using the error backpropagation algorithm and is the most widely used type of neural network [16, 17]. The performance of the classification algorithms was compared in terms of AUC (internal validation), RMSE, R² and MAE. The algorithms showing the best performance were used for total validation.

Single associative mediation effect among environmental factors

In this study, Hayes process (v.3.5) within SPSS 21.0 (SPSS Inc., Chicago, IL, USA) was employed to validate the associative mediation hypotheses regarding a recent study. To assess the mediation effect of PCOS diseases and blood indicators on the predicted variables, a Hayes process moderation analysis was conducted, with 5000 bootstrapping-based resamples chosen. In this context, to emphasize the importance of AMH in the diagnosis of PCOS, this mediation effect model uses the AMH index as an effect factor for the disease. The selected mediating factors are environmental variables that exhibit significant and substantial effect values in the multivariate

stratified analysis. The mediation model structure is as follows: (i) Independent variables: Hormones and serum indicators related to PCOS. (ii) Mediator variables: Air pollutants. (iii) Outcome variable: AMH hormone, which is an important diagnostic marker for PCOS.

Statistical analysis

Statistical analyses were performed using SPSS 21.0 (SPSS Inc., Chicago, IL, USA) and R software (version 4.1.3). We employed univariate non-parametric tests and Spearman correlation analysis to identify factors associated with the prevalence of PCOS. Subsequently, to investigate the association of clinical and environmental factors on PCOS, we performed stratified modeling using logistic regression and WQS regression. Initially, clinical or environmental factors were individually included in the model to calculate their risk effects. Next, both clinical and environmental factors were combined in the model to further explore their effects. The grouped boxplot method was initially applied to generate boxplots, with the mean values of grouped independent variables as the abscissa and logit (PCOS incidence) as the ordinate. The results showed that the scatter points approximately exhibited a linear trend. So, the model here uses multivariate logistic regression, with risk factors selected through the likelihood ratio test (forward: LR) based on maximum likelihood estimation. Additionally, to examine subsequent associative mediation effects, we utilized AMH hormone levels as a diagnostic criterion for PCOS, analyzing the risk effects of blood parameters on AMH levels. To account for age when determining the effects of hormones and environmental exposure on the population, we performed a subgroup analysis based on age, with age groups including 0–20 years and 21–35 years. To establish a predictive model for PCOS, BP neural network algorithms were used.

For this study, multiple interpolation was performed to fill in the missing values, where the parameter $m=5$ and predictive mean matching (pmm) was chosen. Factors influencing PCOS diseases were analyzed separately using univariate analysis and multiple regression. Among univariate analyses, t-tests, non-parametric tests and Spearman analysis were applied respectively. A p-value of <0.05 for the two tails was deemed scientifically significant. This study used Hayes process (v.3.5) by SPSS 21.0 to validate the associative mediation hypotheses for a recent study. For the direct relationship of PCOS diseases and blood parameters, multiple regressions were used in R software (version 4.1.3), and to check the associative mediation effect of the different environmental factors on these predicted variables, Hayes process moderation analysis was used and 5000 bootstrapping-based resamples were chosen.

Results

Comparison of PCOS group with clinical and environmental characteristics

From Table 1, the general and clinical characteristics were statistically significant in the group with PCOS diseases, by age, sexual function indicators, thyroid function indicators, and liver function indicators. Among the sexual function indicators, PCOS patients show significantly higher levels of LH, PRL, T, and AMH compared to the control group. In terms of thyroid function indicators, PCOS patients exhibit significantly lower levels of TGAb than the control group. Regarding liver function indicators, with the exception of the PA indicator, the remaining liver function levels significantly associated with PCOS are higher than those in the control group. Table S1 showed the correlation of environmental indicators among the PCOS diseases patients. It was concluded that all other factors show a positive correlation (coefficient >0), except for the mean temperature.

Multifactorial regression and features analysis

Table 2 shows the results of the overall and stratified regression of PCOS patients, where $OR > 1$ shows a positive effect, including T, AMH, $PM_{2.5}$, Air pressure and Alb, while negative effects included age, FSH and PA. After adjusting for confounding factors, the overall AMH level in the PCOS patient population is 1.133 times that of the non-affected population, with a 95% confidence interval of (1.077, 1.192). In the age group of 21–35 years, the air pressure and albumin levels in the PCOS patient population are 1.060 and 1.098 times those of the non-affected population, respectively, with 95% confidence intervals of (1.028, 1.093) and (1.002, 1.204), respectively. The results from WQS regression modeling shown in Figure S1 indicate that, prior to adjusting for confounding factors, the blood factors positively correlated with the incidence of PCOS are primarily hormonal indicators and liver function indicators. The hormonal indicators include T and AMH, while the liver function indicators include TP, A/G, and TBA. Conversely, the blood factors negatively associated with the incidence of PCOS primarily consist of hormonal indicators, including FSH and SHBG, along with TGAb and PA (Figure S1 A and B). Serum hormones and liver and kidney function indicators were associated with a risk effect on the onset of PCOS.

After adjusting for environmental factors, the blood factors positively correlated with the incidence of PCOS are primarily hormonal indicators and liver function indicators. The hormonal indicators include T and AMH (factor weights of 0.080 and 0.319, respectively), while the liver function indicators are TP, A/G, ALP, and TBA (factor weights of 0.114, 0.107, 0.064 and 0.068, respectively). Among the environmental factors, the primary positive

Table 1 Differentiation of General, Sexual, Thyroid, liver and renal function indicators in women with PCOS and controls

Variables	PCOS (mean (SD) / median (IQR))		P
	Case (n = 178)	Control (n = 206)	
Age	25(22, 29)	33(26, 41)	< 0.0001
FSH(mIU/mL)	5.68(4.64, 6.69)	6.47(4.98, 9.14)	< 0.0001
LH(mIU/mL)	10.60(6.77, 16.48)	7.38(3.98, 14.16)	0.002
PRL(ng/mL)	12.68(9.13, 18.14)	11.45(7.95, 16.19)	0.012
P(ng/mL)	0.56(0.34, 1.05)	0.49(0.28, 0.86)	0.089
E2(pg/mL)	41.94(32.12, 62.08)	39.03(28.84, 73.40)	0.472
T(ng/mL)	0.72(0.61, 0.89)	0.53(0.36, 0.69)	< 0.0001
SHBG(nmol/L)	26.55(15.30, 46.60)	35.85(20.50, 61.40)	0.005
AMH(ng/mL)	8.91(6.11, 13.02)	3.84(1.37, 6.64)	< 0.0001
FT3(pmol/L)	4.24 ± 0.64	4.19 ± 0.68	0.489
FT4(pmol/L)	13.42(12.46, 14.37)	13.14(12.24, 13.98)	0.114
TSH(uIU/mL)	1.64(1.17, 2.50)	1.72(1.15, 2.39)	0.965
TPOab(IU/mL)	1.20(0.39, 3.27)	1.12(0.47, 5.08)	0.2
TGAb(IU/mL)	1.57(0.98, 3.33)	1.74(1.22, 9.21)	0.015
TP(g/L)	73.08 ± 4.40	71.96 ± 5.18	0.023
Alb(g/L)	45.1(43.9, 47.3)	44.6(42.3, 46.5)	0.003
A/G(1)	1.6(1.5, 1.8)	1.5(1.3, 1.7)	< 0.0001
AST(U/L)	17(15, 20)	16(13, 22)	0.157
ALP(U/L)	62(48, 73)	58(47, 70)	0.019
GGT(U/L)	17(12, 30)	17(12, 26)	0.962
PA(g/L)	0.22(0.19, 0.25)	0.23(0.21, 0.26)	0.013
TB(μmol/L)	9.2(7.1, 12.0)	9.4(7.4, 11.7)	0.69
CB(μmol/L)	2.95(2.10, 3.90)	2.8(2.0, 3.5)	0.126
UCB(μmol/L)	6.3(4.5, 8.6)	6.6(4.8, 9.0)	0.379
TBA(μmol/L)	2.8(1.7, 4.6)	2.2(1.4, 3.8)	0.004
MAO(U/L)	5.28(4.18, 7.47)	5.26(3.91, 7.09)	0.512
urea(mmol/L)	4.32(3.6, 5.0)	4.15(3.26, 5.06)	0.143
Cr(μmol/L)	54.3(49.0, 59.6)	55.5(48.4, 61.5)	0.424
UA(μmol/L)	341(279, 403)	348(274, 429)	0.783

Bold font indicates statistical significance at the 0.05 level

Independent-samples t-test or Mann-Whitney U test were utilized for two-way comparisons

Abbreviations: SD Standard deviation, IQR Interquartile range, FSH Follicle-Stimulating Hormone, LH Luteinizing Hormone, PRL Prolactin, P Progesterone, E2 Estradiol, T Testosterone, SHBG Sex Hormone-Binding Globulin, AMH Anti-Müllerian Hormone, FT3 Free Triiodothyronine, FT4 Free Tetraiodothyronine, TSH Thyroid-Stimulating Hormone, TPOa Thyroid Peroxidase Antibody, TGAb Thyroglobulin Antibody, TP Total Protein, Alb Albumin, A/G Albumin/Globulin Ratio, AST Aspartate Aminotransferase, ALP Alkaline Phosphatase, GGT γ-Glutamyl Transferase, PA Prealbumin, TB Total Bilirubin, CB Conjugated Bilirubin, UCB Unconjugated Bilirubin, TBA Total Bile Acid, MAO Monoamine Oxidase, Cr Creatinine, UA Uric Acid

effect factor is air pressure (factor weight of 0.146). Conversely, the blood factors negatively associated with the incidence of PCOS mainly consist of hormonal indicators and environmental factors. The hormonal indicators include FSH and SHBG (factor weights of 0.106 and 0.099, respectively), while the environmental factors are PM₁₀, NO₂, and mean temperature (factor weights of 0.066, 0.078, and 0.263, respectively). Additionally, TGAb and PA are included (factor weights of 0.202 and 0.090, respectively) (Figure S1 C and D). Environmental factors were associated with a risk effect on the onset of PCOS.

According to Table S2, after adjusting for factors, we found that the risk effects of blood indicators on AMH levels are as follows: there are three thyroid function indicators, namely FT4, TSH and TPOab, with risk effects of 0.257, 0.185 and −0.146, respectively. There are two liver

function indicators, CB and Alb, with risk effects of 0.103 and 0.180, respectively. Additionally, there are two environmental factors, O₃ and air pressure, with risk effects of 0.116 and 0.226, respectively.

Algorithm selection and prediction effect

According to Table S3, the PCOS disease modeling was configured with three hidden layers, with 6, 5 and 5 nodes, respectively. In the overall population, under eight different functions and algorithms, the neural network model using a ce loss function, logistic activation function, and rprop+ algorithm demonstrated good predictive performance (AUC(test, train) = (0.864, 0.992), R²(test, train) = (0.342, 0.910)). In the Age 0–20 group, the model employing a sse function, tanh activation function, and rprop+ algorithm also showed favorable

Table 2 Multivariate logistic regression analysis of risk factors of PCOS diseases among different groups

Stratification	Predictor Variables	Model 1		Model 2	
		Exp(β)(95%CI)	P	Exp(β)(95%CI)	P
Total	Age	0.910(0.875, 0.947)	<0.0001	0.888(0.852, 0.925)	<0.0001
	FSH	0.908(0.832, 0.991)	0.03	0.917(0.841, 0.999)	0.046
	T	2.937(1.031, 8.365)	0.044	-	0.094
	AMH	1.130(1.074, 1.188)	<0.0001	1.133(1.077, 1.192)	<0.0001
	PM _{2.5}	1.008(1.000, 1.016)	0.049	-	0.167
	Air pressure	1.054(1.027, 1.081)	<0.0001	1.067(1.037, 1.098)	<0.0001
Age 0–20 years	PRL	1.260(0.991, 1.601)	0.059	1.260(0.991, 1.601)	0.059
	AMH	1.596(1.130, 2.255)	0.008	1.596(1.130, 2.255)	0.008
Age 21–35 years	AMH	1.159(1.093, 1.229)	<0.0001	1.138(1.071, 1.210)	<0.0001
	PA	0.001(0, 0.314)	0.021	0.000(0.000, 0.048)	0.005
	Air pressure	0(0, 0)	<0.0001	1.060(1.028, 1.093)	<0.0001
	Alb	-	0.066	1.098(1.002, 1.204)	0.045

Model 1: single clinical or environmental indicators for univariate analyses

Model 2: combined clinical and environmental indicators for multivariable analyses

Bold font indicates statistical significance at the 0.05 level

No effect estimates indicated that the factors were not incorporated into the model in stepwise regression

Abbreviations: T Testosterone, FSH Follicle-stimulating hormone, PRL Prolactin, AMH Anti-Müllerian hormone, PA Prealbumin, Alb Albumin

predictive outcomes ($AUC(\text{test, train}) = (0.833, 1)$, $R^2(\text{test, train}) = (0.454, 1)$). For the Age 21–35 group, the neural network model utilizing an sse loss function, logistic activation function, and sag algorithm exhibited good predictive performance as well ($AUC(\text{test, train}) = (0.727, 0.966)$, $R^2(\text{test, train}) = (0.066, 0.898)$). The BP neural network model demonstrated good predictive performance for the onset of PCOS.

Subsequently, the validation of the optimal models for the overall population and each subgroup is illustrated in Fig. 1. In Fig. 1A, the prediction error for PCOS in the overall population is shown to be 0.031. In Fig. 1B, the prediction error for the Age 0–20 group is observed to be 0, while in Fig. 1C, the prediction error for the Age 21–35 group is 0.061. The cutoff values are within the range of 0.03 to 0.09. The predictive effect varied across different age groups.

Analysis of the mediating effect

Table 3 presents the results of the associative mediation pathways. For the associative mediation effect results, we describe the findings in the order of total effect, direct effect, and indirect effect. In Model 1, the total effect of testosterone (T) on AMH levels mediated by air pressure is 8.139, with a 95% confidence interval of (6.320, 9.959). Both the direct and indirect effects are statistically significant, with the indirectly calculated effect value from Bootstrap sampling being 0.304 and a 95% confidence interval of (0.024, 0.725), accounting for 3.74%.

Figure 2A demonstrates that air pressure has a significant positive mediating effect in the association between testosterone (T) and PCOS ($\beta = 0.3951$). In Model 2, the total effect of follicle-stimulating hormone (FSH) on AMH levels mediated by air pressure is -0.070, with a 95% confidence interval of (-0.106, -0.033). Both direct

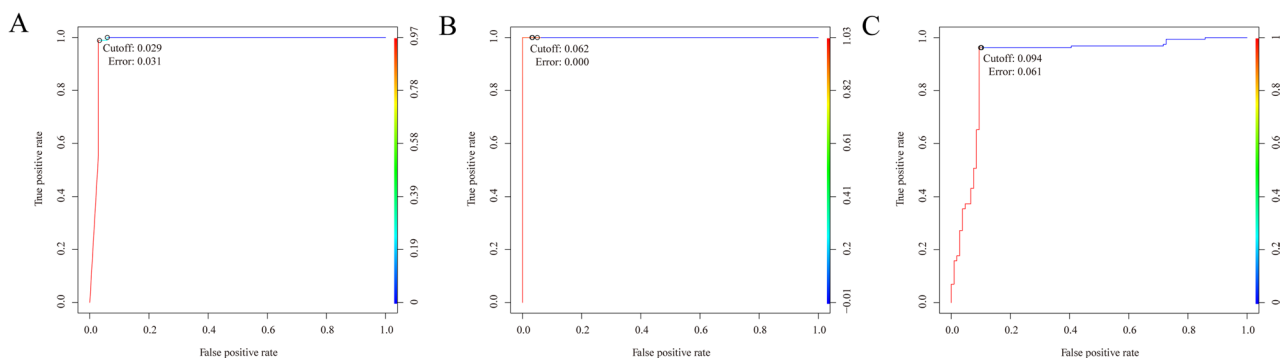


Fig. 1 ROC curve for the best performance in different back-propagation (BP) neural network models. **A** Total group. **B** Age 0–20 years group. **C** Age 21–35 years group

Table 3 Results of the mediation analysis with different factors

Model	Items	Effect	SE	t	P	Boot [LLCI, ULCI]
1	Total effect	8.139	0.925	8.796	0	[6.320, 9.959]
	Direct effect	7.835	0.922	8.498	0	[6.022, 9.648]
	Indirect effect	0.304	0.182			[0.024, 0.725]
	Indirect proportion	3.74%				
2	Total effect	-0.070	0.019	-3.768	0.000	[-0.106, -0.033]
	Direct effect	-0.062	0.019	-3.323	0.001	[-0.098, -0.025]
	Indirect effect	-0.008	0.004			[-0.018, -0.001]
	Indirect proportion	11.89%				
3	Total effect	0.636	0.231	2.759	0.006	[0.183, 1.090]
	Direct effect	0.558	0.232	2.404	0.017	[0.102, 1.015]
	Indirect effect	0.078	0.044			[0.003, 0.174]
	Indirect proportion	12.30%				

Model 1: T→AMH (Mediation: Air pressure)
Model 2: FSH→AMH (Mediation: Air pressure)
Model 3: TSH→AMH (Mediation: PM_{2.5})
Bold font indicates statistical significance at the 0.05 level
Abbreviations: LLCI Lower Limit of Confidence Interval, ULC, Upper Limit of Confidence Interval

and indirect effects are statistically significant, with the indirectly calculated effect value from Bootstrap sampling being -0.008 and a 95% confidence interval of $(-0.018, -0.001)$, accounting for 11.89%. Figure 2B shows that air pressure has a significant negative mediating effect in the association between FSH and PCOS ($\beta=-0.1668$). In Model 3, the total effect of thyroid-stimulating hormone (TSH) on AMH levels mediated by PM_{2.5} is 0.636, with a 95% confidence interval of $(0.183, 1.090)$. Only the indirect effect is statistically significant, with the indirectly calculated effect value from Bootstrap sampling being 0.078 and a 95% confidence interval of $(0.003, 0.174)$, accounting for 12.30%. Figure 2C indicates that PM_{2.5} has a significant positive mediating effect in the association between TSH and PCOS ($\beta=0.1226$). Environmental factors may modulate the association of various hormones (including hormones related to AMH) on the risk of PCOS.

Discussion

In terms of sexual function indicators, PCOS patients exhibited significantly higher levels of LH, PRL, T, and AMH compared to the control group. This is attributed

to elevated LH and testosterone levels, which interfere with normal follicular development, leading to the inability of follicles to mature and ovulate, resulting in multiple immature follicular cysts. Research indicates that 10%-30% of PCOS patients experience mild elevations in prolactin levels. Although the PRL levels in PCOS patients were higher than those in the control group in this study, prolactin can return to normal levels as the levels of testosterone and other androgens gradually decrease. A study conducted in the Netherlands found that serum AMH levels should be considered a biomarker for PCOS [18]. Some studies have also confirmed that when transvaginal ultrasound examination is not feasible, AMH can serve as a useful alternative for diagnosing PCOS, aiding in the identification of women with typical PCOS characteristics [19]. Therefore, this study uses AMH levels as a biomarker for PCOS. Although this indicator is not universally accepted as a diagnostic standard, it still holds clinical significance for the diagnosis of PCOS and as a biomarker, as confirmed by a study in Portugal [20]. This finding is consistent with the results of this study. Therefore, this study integrates literature and recent expert consensus to utilize AMH levels as a diagnostic criterion

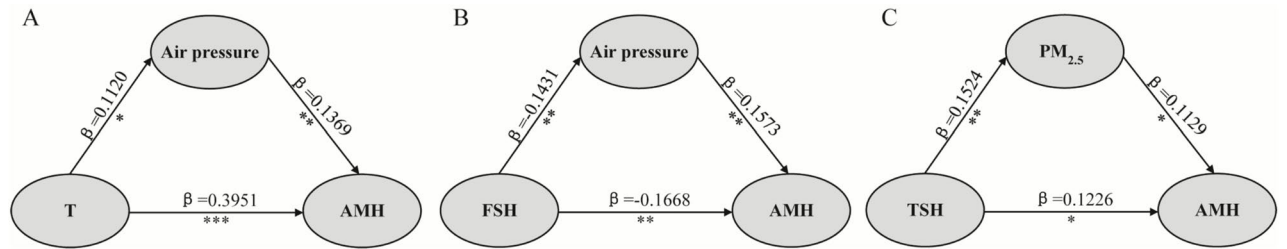


Fig. 2 Path diagram of the mediation analysis of environmental risks on the relationship between blood indicators and AMH. The graphs in (A–C) represented the mediating role of air pressure, air pressure and PM_{2.5}, respectively. Abbreviations: T, Testosterone; FSH, Follicle stimulating hormone; TSH, Thyrotropin; AMH, Anti-Müllerian hormone

for PCOS. Thyroid function factors exhibit varying degrees of expression in the endometrium, particularly during the ovulatory phase.

Research has shown that TSH receptors are widely expressed in human oocytes, cumulus cells, and ovarian granulosa cells. Hypothyroidism can lead to ovulatory dysfunction and menstrual irregularities, and young patients may sometimes exhibit polycystic changes in the ovaries, along with weight gain, making diagnosis of PCOS more likely [21]. This is similar to the finding in this study that PCOS patients exhibited low levels of TGAb expression. This study also found that liver function parameters were significantly elevated in the majority of PCOS patients, further confirming that PCOS can exacerbate hepatic steatosis and liver dysfunction [22]. In the stratified analysis, we found that PCOS in individuals under the age of 25 is more closely associated with sex hormones, while the risk in mature women is more reliant on liver function parameters. This leads us to hypothesize: Younger individuals often have irregular lifestyles, leading to instability in sex hormone levels, which can easily result in dysfunction of the uterus and ovaries due to hormonal fluctuations. In contrast, women over 25 may experience liver dysfunction due to certain occupational factors or life stressors, which further affects the ovaries and contributes to abnormal polycystic changes.

In clinical studies, multiple pieces of literature have indicated that environmental factors such as weather and air pollution significantly association endocrine and metabolic diseases. However, there is currently no research demonstrating an association with PCOS. This study found that high atmospheric pressure increases the risk of developing PCOS and plays a positive mediating role in the influence of sex hormones on PCOS. Although the underlying mechanisms remain unclear, this provides some evidence for a potential link between meteorological conditions and adverse health outcomes in women. A team from the University of California validated in animal models that sub-chronic exposure to PM_{2.5} depletes ovarian reserve by increasing the recruitment of primordial follicles into the growing pool and enhancing the apoptosis of growing follicles [23]. Additionally, a recent birth cohort study found that high concentrations of PM exposure are associated with maternal thyroid dysfunction during early pregnancy [24]. This is consistent with our findings, where the PM_{2.5} levels in the PCOS patient population were 1.008 times higher (95% confidence interval: (1.000, 1.016)) than those in the non-affected population. However, the association size is small. Even if the association size is small, the long-term association of air pollution on public health cannot be ignored. Even though the association on individual cases

may be weak, an increase in air pollution levels across the population may lead to widespread negative health association on society. Therefore, reducing air pollution remains an important public health measure to protect women's health, particularly in preventing diseases such as PCOS. Furthermore, our study concluded that PM_{2.5} may exacerbate the association of TSH hormone on the risk of developing PCOS, providing a reference for future validation in animal models. However, the environmental factor levels selected in this study were derived from the monitoring station nearest to the patient population. Since the exposure range may be limited, and unmeasured confounding factors (such as urban-rural differences or seasonal associations) may influence the results, there may be some error and limitations.

Several studies have shown that in women with PCOS, the levels of total testosterone and AMH in follicular fluid are higher, and it has been concluded that the molecular characteristics of PCOS follicular fluid may be related to cellular microenvironments such as glycolysis and mitochondrial function [25]. Additionally, hypothyroidism may lead to ovulatory dysfunction and irregular menstruation [26]. This study also found that a high-pressure environment could increase AMH hormone levels by affecting testosterone, while PM_{2.5} could decrease AMH levels by influencing TSH. The possible mechanisms are as follows: High pressure may affect hormone secretion by activating the neuroendocrine axis (particularly the hypothalamic-pituitary-adrenal axis) and the sympathetic nervous system (SNS), thereby influencing testosterone and AMH levels. It may also affect the ovarian microenvironment, where stress-induced oxidative stress and immune response changes could lead to ovarian dysfunction, causing an increase in AMH levels. Furthermore, long-term exposure to PM_{2.5} can trigger a systemic inflammatory response, stimulating the immune system and activating cytokines in the body. This inflammatory response may influence thyroid function, leading to elevated TSH levels. Hypothyroidism may result in decreased ovarian reserve, as reflected by a decrease in AMH. AMH is commonly used to assess ovarian reserve, and its decrease could be an early indicator of ovarian function decline [27].

To highlight the importance of AMH hormone in the diagnosis of PCOS, we used AMH as a predictor in the associative mediation model. The selected mediators were environmental factors that demonstrated significant and substantive association values in the multiple stratified analysis, including both PCOS risk factors and AMH-level modifiers. This approach improves diagnostic accuracy and supports the established role of AMH in PCOS diagnosis.

This study explored its objectives using various statistical models. In terms of the predictive associations of environmental factors and blood test indicators on PCOS using neural network models, different modeling algorithms yielded varying results across different groups. This finding has been well validated in multiple clinical studies utilizing neural network models. However, due to the limitations of our sample size and the weak external validity of single-center validation, we did not employ other machine learning algorithms for comparison and validation. Additionally, due to the insufficient data volume and the lack of confounding factors in this study, the results exhibit some implausibly favorable outcomes (such as zero prediction error in subgroup analysis), indicating the limitation of overfitting. This highlights the need for future research to include sufficiently large, multi-center cohorts for systematic analysis to enhance the predictive performance of complex models. Furthermore, in terms of predictive indicators, this study incorporated the AUC on top of the R^2 , but lacked performance in classification metrics (such as accuracy, precision, recall, and F1 score). This is mainly because this study focused on analyzing the prediction performance of the same model with different algorithms. Future research should consider comparing various machine learning models using multiple classification metrics. The specific manifestations are as follows: Retrospective studies and single-center studies that rely on historical data from a specific population may reduce representativeness, affecting the generalizability and external validity of the results. The selection of study subjects may be subject to certain biases, particularly during the matching process between the case and control groups, where systematic differences may exist. Additionally, an insufficient sample size may decrease the statistical power of the study, leading to false-negative results (i.e., failing to detect an actual association) and an increased risk of overfitting. Furthermore, since diseases such as obesity can increase the risk of PCOS, this study lacks investigation into certain disease factors including BMI and metabolic indicators. It will be difficult to determine whether the observed differences (e.g., hormone levels or liver function) are attributed to PCOS itself or pre-existing weight differences. This highlights the need for future research to account for environmental factors while also incorporating confounding factors to enhance the reliability of the study findings. We also hope that future researchers will further investigate the mediating associations of environmental factors on the risk of PCOS by establishing animal models and conduct multicenter studies to validate the predictive associations of these environmental factors. This study selected a group exposure associations for environmental exposure, which to

some extent overlooks the individual-level exposure, and therefore cannot be fully considered as causal inference. This limits the accuracy of linking the change in PCOS risk with environmental pollutants, a common limitation in many environmental epidemiology studies. However, the aim of this study was to preliminarily explore the environmental sensitivity factors for PCOS, so as to lay a foundation and basis for future, more precise detection of exposure areas within patients' residential locations. Additionally, conducting multi-center comparisons across different cities or in areas with higher exposure levels may yield more pronounced associations.

At the same time, we will conduct external validation of the independent dataset and its predictive model in the future, which is crucial for evaluating the generalizability of the predictive model. We also advocate for future multi-center, prospective studies utilizing larger sample sizes. The association of environmental factors (such as meteorological conditions, pollutants, etc.) on human health has been widely discussed, but how environmental factors affect hormone levels or ovarian function in PCOS remains unclear. This study attempts to explore the intersection of meteorology and environmental pollution, providing new scientific insights. In public health policy formulation, this study offers new evidence for the relationship between air pollution and health, particularly its association on women's health. If the results are further validated, governments and public health institutions may pay more attention to the potential threats of environmental pollution on women's health (especially reproductive health), leading to stricter air quality management and pollution control policies. Air pollution has become a global health issue, and this study suggests a possible association between $PM_{2.5}$ and PCOS, which could encourage cities and industrialized regions to adopt more active air purification measures. Regarding women's health management, understanding the potential association of environmental pollution on PCOS could provide new perspectives for early disease screening and intervention. For instance, in high-pollution areas, women could receive more health education and preventive measures, such as regular endocrine level checks, attention to environmental factors, and lifestyle adjustments. Based on the potential association of environmental pollution on PCOS, future health guidance could offer more specific help for women to avoid harmful environmental exposures (such as $PM_{2.5}$, low air pressure, etc.) and provide personalized advice based on individual health conditions. Healthcare providers can focus more on the association of environmental factors on PCOS in female patients in high-pollution areas, thereby offering more comprehensive diagnosis, treatment and care.

Conclusion

Significant elevations in levels of AMH and Alb were detected in women with PCOS. The back-propagation (BP) neural network demonstrated good PCOS predictive performance for the models mediated by environmental factors (PM_{2.5}, air pressure). This suggests that these factors may probably exacerbate the associations of sexual function (FSH, PRL, T, AMH) and liver function indicators (Alb, PA) on the risk of developing PCOS. Our results support a potential association between environmental factor exposure and the consequences of PCOS in women.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13048-026-01981-7>.

Supplementary Material 1: Figure S1 WQS model regression index weights for blood indicators and PCOS diseases in women. Model 1: single clinical or environmental indicators for univariate analyzes. Model 2: combined clinical and environmental indicators for univariate analyzes. Table S1 Spearman's correlation of PCOS diseases with meteorological and polluted parameters. Table S2 Multivariate logistic regression analysis of risk factors for AMH level. Table S3 Model performance metrics of different functions and algorithms among different groups in BP neural network.

Authors' contributions

WH: Software, Conceptual, Methodology, Formal analysis, Investigation, Resources, Writing-original draft, Writing-review & editing. JD: Software, Conceptual, Methodology, Formal analysis, Investigation, Writing-original draft, Writing-review & editing. YY: Software, Conceptual, Methodology, Formal analysis, Investigation, Writing-original draft, Writing-review & editing. PZ: Conceptualization, Methodology, Formal analysis, Writing-original draft, Writing-review & editing, Supervision. GZ: Methodology, Software, Writing - original draft, Visualization. PM: Methodology, Software, Writing - original draft, Visualization. JG: Methodology, Software, Writing - original draft, Visualization. YW: Conceptualization, Methodology, Formal analysis, Writing-review & editing, Funding acquisition, Supervision.

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Data availability

The data are not publicly available due to restrictions that could compromise the privacy of research participants. Requests to access the datasets should be directed to wangying_shengjing@163.com.

Declarations

Ethics approval and consent to participate

The studies involving humans were approved by The Ethics Committee of the Shengjing Hospital of China Medical University (ID: 2024PS123K). The studies were conducted in accordance with the Declaration of Helsinki, local legislation and institutional requirements. The participants provided their written informed consent to participate in this study. Written informed consent was obtained from the individual(s) for the publication of any potentially identifiable images or data included in this article.

Consent for publication

All authors have consented to publication of this research.

Competing interests

The authors declare no competing interests.

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